AUTOMATED FINANCIAL INTELLIGENCE PIPELINE

Product Research and Technical Scope Document

Proposed early-stage AI fintech project focused on data understanding, verified analytics, and explainable financial insights.

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Executive Summary

This document defines a proposed Automated Financial Intelligence Pipeline. The system is intended to accept financial data, understand its structure, apply deterministic financial and statistical analysis, and present verified findings in a context-aware dashboard. A small AI model is a planned downstream explanation layer, not the primary calculator.

Key principle: deterministic computation produces the facts; AI communicates the facts.

The MVP deliberately focuses on personal-finance transaction data in CSV and Excel/XLSX formats. It will calculate core metrics, identify trends and basic anomalies, produce structured insights, and display them in a dashboard. DOCX, PDF/OCR, business statements, investments, forecasts, and recommendations are future scope.

Contents

1. Project Context and Intended Workflow

The user uploads financial information. The proposed system identifies the file type, extracts data, cleans and normalizes it, checks quality, detects schema and financial context, calculates applicable metrics, performs statistical analysis, detects trends/patterns/anomalies, identifies drivers, creates structured findings, and then optionally generates AI explanations.

Pipeline

RAW DATA → EXTRACTION → CLEANING → QUALITY CHECK → SCHEMA UNDERSTANDING → FINANCIAL ANALYSIS → STATISTICS → TRENDS / PATTERNS / ANOMALIES → STRUCTURED FINDINGS → AI EXPLANATION (PLANNED) → DASHBOARD

Critical AI Boundary

Python, Pandas, NumPy, statistics, and financial rules should calculate and verify numerical findings. For example: income INR 60,000; expenses INR 48,000; savings INR 12,000; savings rate 20%; shopping change +58%. The AI layer may turn only those verified findings into readable language.

Why this matters	Result
Reliability	Numerical results are reproducible and testable.
Debugging	Each calculation and transformation has a visible stage.
Hallucination control	Language generation follows evidence rather than raw files.
Finance validation	Domain members can review rules and findings.

2. Existing Projects and Research References

The following are references for individual ideas. They are not claimed as components of this project, and their functionality is not attributed to the proposed system.

Reference	What it demonstrates	Learning / boundary
Personal Finance Dashboard vi nzalfaro/personal-finance-dashboard	Python, Pandas, Plotly, Streamlit and SQL for expenditure analytics, budgeting, inflows/outflows and patterns.	Useful dashboard reference. Primarily tracking, analytics and visualization; our proposed system aims to add automated interpretation, trends, anomalies and AI explanation.

Reference	What it demonstrates	Learning / boundary
Streamlit Exploratory Analysis camilasbraz/streamlit-exploratory-analysis	CSV/Excel upload and automated EDA profiling: descriptive statistics, missing values, outliers, correlations, distributions, trends and patterns.	Shows generic automated data understanding. It does not establish finance-specific schema interpretation or financial intelligence.
FG-Data-Profiling Data-Centric-AI-Community/fg-data-profiling	Automated Pandas/Spark profiling: type inference, duplicates, univariate/multivariate analysis, correlations, time-series, seasonality, alerts and reports.	Repository identifies itself as renamed successor to ydata-profiling/data-profiling. General-purpose profiling, not a financial intelligence engine.
Financial Statement Analysis tenPro4/pandas_financial_statement	Revenue, expenses, profit, margins, balance sheet, cash flow, trends and segments.	Identifies possible future financial metrics. It does not mean our system already implements them.
S&P 500 Stock Analysis Dashboard Chennakeshav2003/SP500-stock-analysis-dashboard	Returns, growth, risk, volatility, correlation, performance and market trends.	Not our target application; a reference for future financial analytics and visualization techniques.
Personal Finance Tracker DoshiHarsh/Personal-Finance-Tracker	Income, expenses, categories, transactions, budgets and balances.	Reference for records and categorization. Our intended workflow analyzes existing uploaded data rather than manual maintenance of every record.

3. Research Gap / Opportunity

The references demonstrate complementary parts: personal-finance analytics and visualization, general EDA, data profiling and quality analysis, financial-statement metrics, market analytics, and financial record management. The proposed system aims to combine selected ideas into a finance-context-aware pipeline.

This does not claim absolute novelty or that no similar system exists. The proposed differentiation is the traceable sequence from ingestion and quality checks to finance-defined analysis, structured findings, constrained AI explanation, and an analysis-driven dashboard.

Project	Primary purpose	Automation	Financial intelligence	AI explanation	Our extension
Personal Finance Dashboard	Personal finance analytics	Partial	Reference	No stated layer	Automated findings and explanation
Streamlit EDA	Generic EDA	Yes	No	No	Financial-domain layer
FG-Data-Profiling	Profiling / quality	Yes	No	No	Context and financial metrics
Statement / market references	Financial analysis	Reference	Reference	No stated layer	Validated future modules
Proposed system	Automated financial intelligence	Planned	Planned	Planned downstream	Integrated, evidence-first workflow

4. Proposed System and Realistic MVP

User Journey

A user uploads expenses.xlsx. The system extracts data, normalizes fields such as Date, Description, Category, Amount and Income/Expense, checks quality, detects the financial context, runs applicable rules/statistics, and presents verified results. A future AI service converts the findings into constrained explanation cards.

Area	MVP commitment
Inputs	CSV and Excel/XLSX
Target	Personal-finance transaction data
Core fields	Date, description, category, amount, income/expense

Area	MVP commitment
Outputs	Income, expenses, savings, savings rate, category breakdown, monthly trends, top categories, basic anomalies, structured insights and dashboard
AI	Planned explanation of verified findings; model training follows a working analytics pipeline
Deferred	DOCX/PDF/OCR, business statements, investment analytics, forecasts and recommendations

Illustrative Finding Contract

A structured finding should include values, period, supporting fields, the rule or method version, and review status. Illustrative examples - not implemented output or research results - include income 60,000, expenses 48,000, savings 12,000, savings rate 20%, major trends, anomalies, expense drivers and risk flags.

An anomaly means a material deviation from expected historical behavior. It does not automatically mean fraud, an error, or a bad transaction.

5. Financial Analysis Scope and Personalization

Phase 1 / Core	Phase 2 / Advanced - planned
Total income; total expenses; net savings; savings rate; expense distribution; category-wise spending; monthly comparisons; income-versus-expense trends; largest categories; recurring expenses; basic anomalies; period-over-period changes.	Trend detection; seasonality; correlation; expense drivers; cash flow; financial ratios; profitability/margins; business financial analysis; investment performance; risk indicators; forecasting.

Technical interpretation: core metrics are calculated from cleaned records and explicit aggregation rules. Advanced methods require clear input requirements and validation. Correlation shows measures moving together; it does not prove causation.

Finance/business interpretation: core analysis answers where money is going, whether spending is changing, whether income covers expenses, and which activity needs review.

Analysis-Driven Dashboard Generation

Detected context	Dashboard focus
Personal finance	Income, expenses, savings, debt, cash flow and spending patterns
Business financial data	Revenue, COGS, gross profit, operating expenses, net profit, margins and cash flow
Investment data	Portfolio value, returns, allocation, volatility, performance and risk

The dashboard should be dynamic based on detected financial context, not generic. Data structure and validated analysis determine which insight cards and visualizations are relevant.

6. Technical Architecture

USER → FRONTEND → UPLOAD API → FILE PROCESSING → NORMALIZATION + QUALITY CHECKS → SCHEMA / CONTEXT DETECTION → FINANCIAL ANALYSIS ENGINE → STRUCTURED INSIGHT ENGINE → DASHBOARD + REPORTING

Component	Technical responsibility	Business purpose
Frontend	File selection, feedback and dashboard	Simple path to insights
Upload API	Receipt, metadata, type/size checks	Controlled ingestion
File processing	CSV/XLSX parse; DOCX/PDF planned	Usable source records
Normalization	Dates, signs, currencies/formats, labels and quality issues	Comparable and trustworthy data
Schema detector	Identify fields and financial context	Select relevant analysis
Analysis engine	Metrics, rules and aggregations	Traceable financial facts
Insight engine	Trends, outliers, patterns and drivers	Prioritize meaningful changes
AI service - planned	Constrained text from findings	Readable explanation
Storage if needed	Secure data/findings/rule versions	Persistence, privacy and auditability

Security is required before production: protected transport, restricted access, retention and deletion rules, safe logging, and a storage design appropriate for sensitive financial information.

7. AI/ML Scope and Team Responsibilities

Proposed AI Scope

The small AI model is a future design decision, to be explored in Google Colab only after deterministic analysis creates stable structured findings. Options may include a small instruction-tuned language model, compact transformer fine-tuning, or parameter-efficient fine-tuning such as LoRA/QLoRA. No completed training or final model choice is assumed.

AI input	AI output
Metric, change, period, category, severity, supporting statistics, finding and allowed wording constraints	Human-readable explanation or insight card that remains faithful to those findings

Financial Decision Support vs Financial Advice

The initial system is for analytics, insights, monitoring, reporting and decision support. It is not professional financial advice, guaranteed investment recommendations, or guaranteed profit/loss predictions. Any future recommendation layer requires finance-domain validation and careful framing.

Finance-domain team	Technical team	AI layer
Define metrics/ratios; validate interpretations; define thresholds; decide useful insights; review findings.	Ingestion, extraction, cleaning, transformation, analytics, statistics, API/backend, dashboard, security and testing.	Summarization, explanation and natural-language generation only; no primary calculations or unverified advice.

8. Development Roadmap

Phase	Objective	Expected output	Success criterion
1	Requirements	MVP contract and finance definitions	Scope/exclusions agreed
2	Sample data	Permissioned test data and edge cases	Coverage documented
3	CSV ingestion	Parser and upload flow	Clear success/failure handling
4	Excel ingestion	XLSX extraction	Required fields extract correctly
5	DOCX extraction	Future prototype	Only after tabular MVP stability
6	Cleaning	Normalized schema and quality report	Traceable transformations
7	Core analytics	MVP metrics engine	Finance review confirms calculations
8	Trends/anomalies	Configured insight methods	Reviewable, useful findings
9	Structured insights	Versioned findings contract	Dashboard works without AI
10	Dashboard	Personal-finance MVP	Users understand results
11	AI training	Colab evaluation experiment	Faithfulness target met
12	AI integration	Guarded explanation service	Text traceable to findings
13	Validation	Test and review results	Release criteria passed
14	Deployment	Controlled pilot/production	Monitoring/support ready

9. Evaluation, Limitations and Risks

Evaluation area	Criteria
Data pipeline	Ingestion success rate; extraction accuracy; cleaning accuracy
Analytics	Metric correctness; trend detection accuracy; anomaly quality; financial-rule correctness
AI	Faithfulness; hallucination rate; explanation quality; readability
Dashboard	Usability; metric relevance; personalization; clarity
End-to-end	Upload → analysis → insight → explanation → dashboard

Known Limitations

Financial data may be incomplete, inconsistent or ambiguously labeled. Conclusions depend on data quality and selected rules. Different terminology can break automatic mapping. AI explanations can still be wrong. The system must show uncertainty, avoid unsupported claims, and protect sensitive data.

Risk	Initial mitigation
Technical variability	Narrow CSV/XLSX MVP and clear unsupported-format feedback
Financial interpretation	Finance review and versioned rules
AI hallucination	Structured-input-only AI and output checks
Privacy	Least access, protected storage/transit, retention/deletion policy

Risk	Initial mitigation
Scope creep	Protect the MVP boundary
Scalability	File limits, async jobs and performance tests

10. Future Scope and What We Should Build First

What We Should Build First

CSV / EXCEL → CLEANING → CORE FINANCIAL METRICS → TREND DETECTION → BASIC ANOMALY DETECTION → STRUCTURED FINDINGS → DASHBOARD → AI EXPLANATION

AI model training should come after the deterministic pipeline has a working, testable structured output. This gives the team an evidence base for evaluating whether generated explanations are accurate and useful.

Future Scope - Planned

Possible later extensions include PDF bank statements, OCR, bank API integrations, multiple accounts, business and investment analysis, forecasting, budget recommendations, cash-flow forecasting, financial-health scoring, advanced anomaly detection, multi-user support, cloud deployment, role-based access, secure document storage, explainable AI, and feedback-driven model improvement.

Final Vision

A user uploads financial data without needing Python, Pandas, Seaborn, Matplotlib, statistics or SQL. The system handles analytical work and returns a financial overview, important metrics, trends, patterns, anomalies, drivers, explanations, visualizations and personalized insights. The first success is a trustworthy, narrow CSV/XLSX personal-finance MVP.

Appendix A. Local Data Sources and Existing Analysis

This appendix records evidence found in the local workspace. The original external publisher and download URL for the local datasets are not documented there. File names such as MCA and XBRL suggest a company-reporting context, but filenames alone are not sufficient evidence to claim an official external source. Before external use, record the source URL, collection date, permissions/license, coverage, and transformation history.

Local asset	Observed contents	Use in scanned code	Relevance
XBRL-XBRL-PL-2025_part1.csv	96,632 rows; 130 P&L/reporting fields, including UID, dates, revenue, expenses, profit, taxes, cost lines, industry and report type.	Primary input for EDA_Project_001.ipynb, Final_Destination.ipynb, check.ipynb and c.py.	High - primary analytical source
MCA_PL_FY2024_25_Clean_Analysis.csv	45,414 rows; 19 selected and derived metrics/ratios.	Created in Final_Destination.ipynb and reloaded for distribution/skewness analysis.	High - derived output, not an independent source
company_master_data_2026-08-25 (1).csv	Company identity fields: CIN, name, date, status, capital, ROC, address, state and classification.	Referenced by SQL inspection/query script; no scanned Python notebook reads it.	Medium - possible future enrichment after validated join
Finance_Data_Company_Name.sql	Database setup and company-master inspection/search queries.	Describes, counts and searches company master records.	Supporting metadata workflow

How the Data Was Used

The main notebook performs data-quality checks, removes fully empty fields, filters rows marked current with reporting dates from 2024-04-01 to 2025-03-31, and retains positive-revenue rows for ratio calculations. It derives profit margin, expense ratio, finance-cost ratio, employee-cost ratio, and R&D intensity; then exports the selected dataset. Other notebook cells inspect missingness, duplicates, distribution, skewness, outliers, finance-cost quartiles and correlation. The smaller check.ipynb and c.py files inspect identity-like P&L fields and company UID examples only.

Data Relationship Flow

RAW XBRL P&L; FILE → QUALITY CHECKS → CURRENT FY 2024-25 FILTER → REVENUE > 0 FILTER FOR RATIOS → DERIVED METRICS → CLEAN ANALYTICAL CSV → DISTRIBUTIONS / OUTLIERS / QUARTILES / CORRELATIONS

The company master file is not joined by the scanned Python code. It may later enrich company identifiers only after finance and data teams validate an unambiguous UID/CIN mapping.

Appendix B. Observed Data Results and Relationships

The following results were calculated from the local XBRL P&L file by analyze_local_finance_data.py. They describe the reviewed company-financial dataset, not the proposed user-upload fintech product.

Measure	Observed result	Interpretation
Raw dataset	96,632 rows; 130 columns	Wide company financial-reporting dataset.
Fully empty fields	16 removed; 114 remain	Empty fields were excluded before analysis.
Missing cells	7,001,669	Material missingness; handle per metric.
Duplicate full rows	0	No full-row duplicates found by this check; not proof of no business duplicates.
Current FY 2024-25	48,068 rows	Existing analysis period selection.
Revenue-positive ratio analysis	45,414 rows; 2,654 excluded	Avoids invalid ratio denominators.
Unique company UIDs	45,414	One UID per retained output row.

Measure	Observed result	Interpretation
Median profit margin	1.81%	Descriptive midpoint, not a benchmark.
IQR margin outlier flags	11,410	Statistical extremes; not fraud/error conclusions.
Profit / loss / zero	35,697 / 12,119 / 252	Supports outcome segmentation.

Observed Relationships

Relationship	Observed statistic	Meaning and caution
Revenue vs expenses	Pearson r = 0.999	Very strong scale association; does not establish efficiency or causation.
Revenue vs profit after tax	Pearson r = 0.001	Near-zero linear association; do not treat as a business conclusion by itself.
Expenses vs profit after tax	Pearson r = -0.002	Near-zero linear association; nonlinear and segmented effects may exist.
Finance-cost ratio vs profit margin	Pearson r = -0.168	Weak negative association; requires industry/size review and is not causal.
Finance-cost burden quartiles	Median margin: Q1 3.04%; Q2 2.08%; Q3 1.62%; Q4 1.32%	Descriptive decline across burden quartiles; finance validation required.

Appendix C. Graphs From Local Analytical Data

Graphs use the 45,414 revenue-positive current FY 2024-25 records. The profit-margin histogram clips the displayed tails only; underlying calculations retain original values. The correlation heatmap is a descriptive Pearson-correlation view and does not demonstrate causation.

Figure 1. Profit Margin Distribution

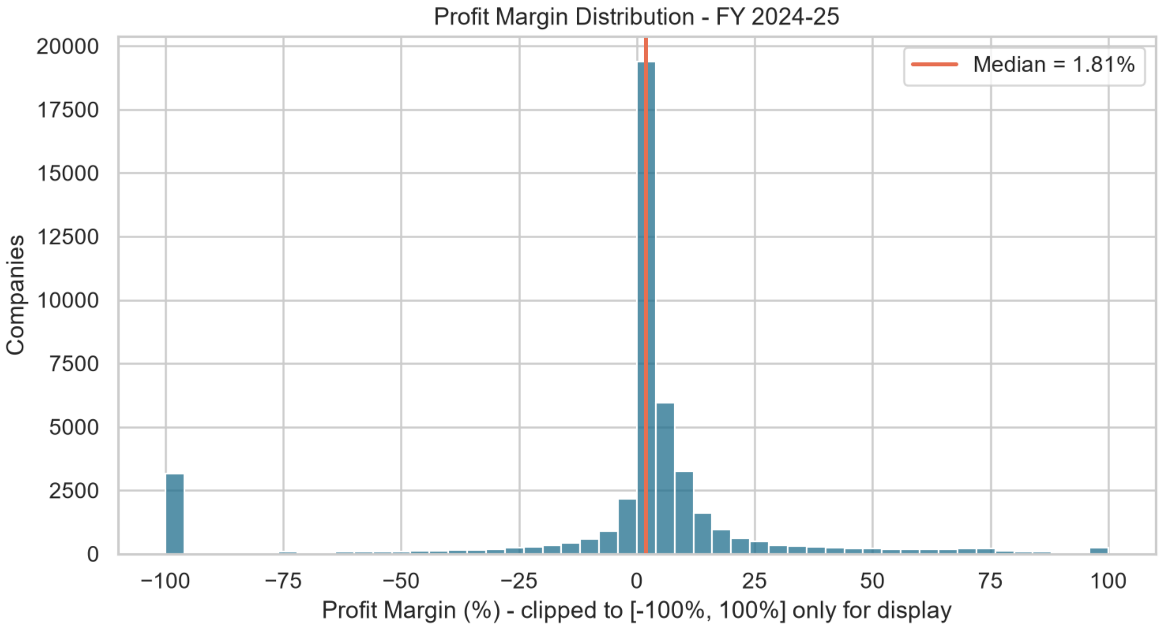


Figure 2. Median Profit Margin by Finance-Cost Burden

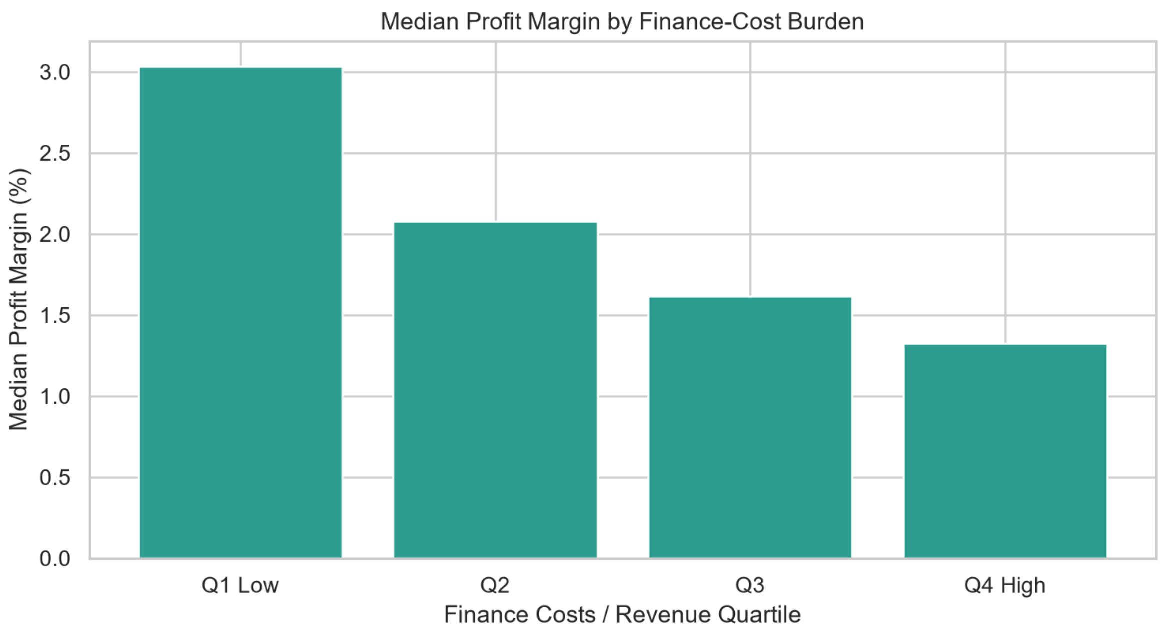
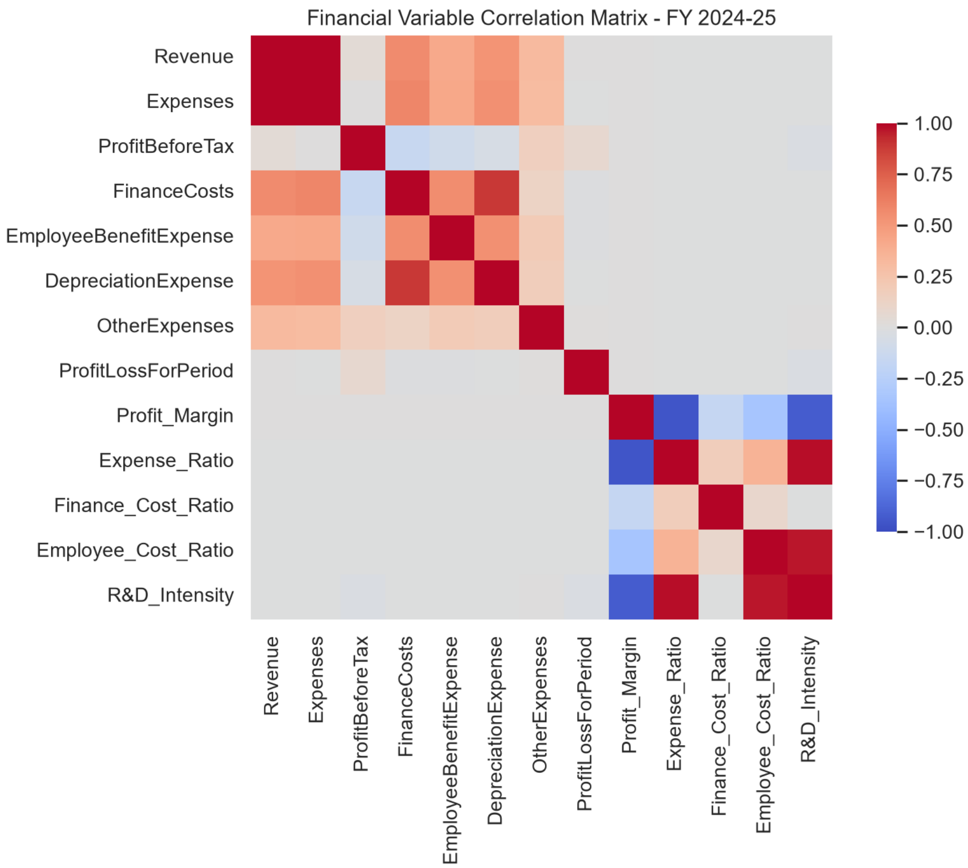


Figure 3. Financial Variable Correlation Matrix



Relevance to the proposed product: this existing work is a business financial-statement analysis prototype, whereas the proposed MVP is transaction-level personal-finance analysis. The reusable lessons are data-quality checks, explicit filters, deterministic ratios, structured outputs, outlier logic and explainable visuals. The contexts and metrics should remain separate.

References

1. https://github.com/vinzalfaro/personal-finance-dashboard
2. https://github.com/camilasbraz/streamlit-exploratory-analysis
3. https://github.com/Data-Centric-AI-Community/fg-data-profiling
4. https://github.com/tenPro4/pandas_financial_statement
5. https://github.com/Chennakeshav2003/SP500-stock-analysis-dashboard
6. https://github.com/DoshiHarsh/Personal-Finance-Tracker

Final quality check: reference systems and the proposed system are clearly separated; advanced work is marked planned; AI is not the calculator; finance ownership is defined; the MVP is constrained; risks and limitations are explicit.